

Fig. 7: A single-side, actual-size PCB layout for eight-channel data acquisition and logging system

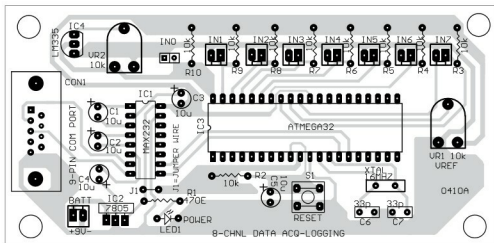


Fig. 8: Component layout for the PCB

PARTS LIST

Semiconductors:

- IC1 - MAX232 RS-232 converter
- IC2 - 7805, 5V regulator
- IC3 - Atmega32 AVR microcontroller
- IC4 - LM335 temperature sensor
- LED1 - 5mm light-emitting diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 470-ohm
- R2 - 10-kilo-ohm
- R3-R10 - 1-kilo-ohm
- VR1, VR2 - 10-kilo-ohm preset

Capacitors:

- C1-C5 - 10 μ F, 16V electrolytic
- C6, C7 - 33pF ceramic

Miscellaneous:

- X₁(TAL) - 16MHz crystal oscillator
- CON1 - 9-pin D-type female connector
- BATT. - 9V PP3 battery
- S1 - Push-on to switch

There are eight input lines (IN0 through IN7) through which analogue inputs are fed into the circuit. The analogue input is converted into digital level by the AVR and transmitted to the PC through the 9-pin, D-type serial com port connector. Here, we have used only three pins of the connector (Rx, Tx and Gnd) for communication with the PC.

PC GUI software

The graphic user interface (GUI) displays on the dashboard the stored data with date and time of logging. This can be useful to analyse the trend of change in temperature. The software dashboard has eight blocks for displaying data of eight different analogue channels.

The GUI software is written in Visual Basic and has MSComm ActiveX controls for communication with the serial port of the PC. It is programmed to scan real-time incoming data from the external hardware. The entire working logic is asynchronous; it doesn't matter which channel has what data. The software can capture the data from a particular channel and put it into an appropriate location in the database. A special protocol is used to synchronise the software with the hardware in order to make the program identify the data and channel number currently active on the serial port. The microcontroller first sends the channel number followed by the current captured data on the channel.

The software is calibrated such that it shows the temperature directly in degree Celsius by multiplying a calibration constant with the incoming analogue voltage.

The software is configured to work with fixed values such as 'com1' for the serial port and '9600' for the baud rate by default. But you can easily configure it to work with different serial ports (like com2, com3 or com4) and baud rates.

The software can save data of the different input channels into a 'daq.mdb' database with time and date of each channel input data. Start/stop buttons are provided to start or stop the logging activity any time by the user. The GUI program output is shown in Fig. 6.

Construction

A single-side, solder-side PCB layout of the circuit for eight-channel data acquisition and logging is shown in Fig. 7 and its component layout in